1. Introduction and Overview

The Concept Development & Design Team will support retrofit designs for first mile retrofits as well as controls & solutions development to be used for future projects.

Key Objectives

1) Develop conceptual designs and requirements to generate RFPs for first mile retrofits and reduce end-to-end design cycle time
2) Develop and pilot internal controls & software solutions to retrofit existing sites and be used as basis for future greenfield projects
3) Explore new product initiatives to drive innovation through pilot programs and graduate technologies into general availability

This document serves as the charter for the [team name] team's success.

2. Mission Statement, and Tenets

Vision: The vision of the [team name] team is to provide innovative hardware and software solutions that accelerates Amazon's competitive advantages to enhance Amazon's existing network as industry leaders in safety, cost, speed, quality, and performance. Our vision is that our concepts and solutions are increasingly modular and advanced, simplifying engagement with vendors and firm engineering, that our first-time designs are pre-engineered solutions leveraging global best practices, but also raising the bar to be the golden standard used by all project types.

Mission: The [team name] team will support the first mile operations network by developing world-class designs evaluating internal and external product offerings that best align with our customer's strategies delivering solutions that enhance ergonomics and safety, ensure robust reliability and equipment maintainability, and drive innovation to support Amazon's growth. The team will partner and develop internal standards and solutions that accelerates the design flywheel, minimizes wasted time through redesign or missing scope, accurately evaluates tradeoffs to maximize ROI while delivering on engineered quality.

Tenets:

Safety of One – The safety of even one individual comes before anything else

Plant Seeds, Harvest Ideas – Be aggressive piloting concepts that could mature to be unfair advantages

Automate to Innovate – Seek out automation of repetitive tasks to free up time for our ourselves and our customers

Sharpen the Scalpel, Not Just the Axe – Dedicated commitment to personal development and learning refines skills

Focus on Inputs for better Outputs – Focus on clear requirements, problem statements, and well-structured designs

Raise the Floor, not Just the Ceiling – Improve baseline solutions, not just cutting-edge to elevate the entire network

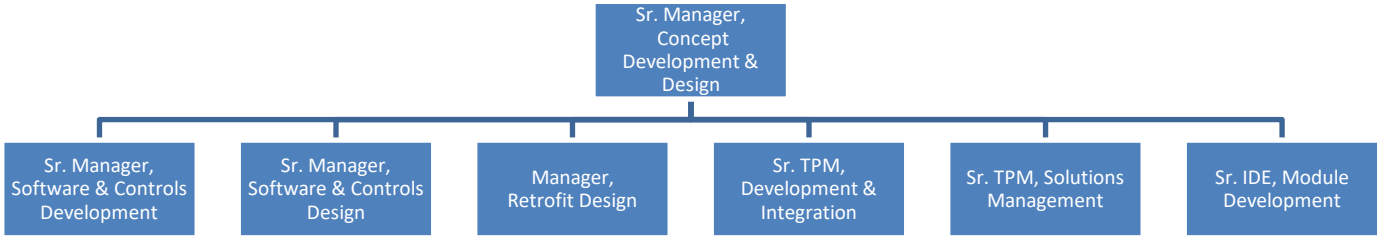
Bias for Action, Built to Last – Move quickly but design with long-term scalability, sustainability, and maintainability

Truth in Data, Truth in Design – Gather robust data, make data-informed decisions, and validate assumptions early

3. Organizational Structure

Team Goals: Our goal is to deliver near firm ready designs minimizing re-engineering time downstream post-RFP and quickly evaluating technology alternatives to maximize ROI for Amazon while meeting design needs. Reduce end to end design cycle time by leveraging internalized standards (Generated from Concept, Seahorse, One-MHS, WWDE, etc.) enabling quicker design to order entry time, cost competitive solutions, and consistent safety & quality built into designs. We will be an active contributor to the retrofit engineering flywheel, proactively identifying engineering opportunities to improve Amazon's network inside of the four walls of an FC. We are force multipliers bringing strong initiatives to light through a collaborative partnership with FTR, RME, Execution Ops Engineering, and Operations. We will develop solutions that helps Amazon scale and we will build solutions with the intent that they can be used not only in a retrofit application, but as the golden standard for future greenfield applications as well.

Team Structure: The team will consist of a combination of System Development Engineers, Industrial Design Engineers, and Technical Program Managers.



Software & Controls Development Team Objectives: This team will develop, pilot, and graduate future industrial controls/software solutions to be used by design teams in vendor-agnostic applications. The long-term vision is for the team to internalize fungible solutions that enhance Amazon's safety, quality, and performance that will be required to retrofit Amazon's network leveraging Amazon's competitive advantage in software and cloud.

Software & Controls Design Team Objectives: This team will integrate software and controls integration requirements and define the functional requirements to be included in RFPs as part of retrofit designs. They will also own full design of projects that require no hardware and are only software/controls retrofits. The long-term vision is to be able to reduce controls scope from vendors and design in more competitive internalized solutions for Amazon.

Retrofit Design Team Objectives: This team will develop conceptual designs and total cost of ownership models for retrofit projects requiring MHE or Storage. The long-term vision is to be able to concept engineer with Amazon maintained standard CAD blocks/modules that accelerates the simulation, technical evaluation, and cost estimation to reduce firm engineering time required.

IC, Development & Integration Objectives: This is a dedicated IC function to support the Software & Controls Development team while also collaborating cross functionally to align strategically and tactically with One-MHS, MSP, Central RME, or other controls development teams. The long-term vision is that this role will be the glue between internal solutions and vendor-based projects.

IC, Solutions Management Objectives: This is a dedicated IC function to support the Retrofit Design team by developing a cross-functional repository of internal and external technologies so that concept engineering can propose the right applications that meet design requirements, ROI, and specifications. The long-term vision is that they are an accelerant for new product initiatives (workcells, vision systems, conveyance solutions, workstations, etc.) that can provide Amazon designs a competitive advantage and accelerates decision making by enabling proper technology tradeoffs, lead time & cost estimate considerations, and equipment quality & constraints.

IC, Module Development Objectives: This is a dedicated IC function to support the Retrofit Design team by developing standardized modules with application standards natively built-in to minimize RFP and firm engineering time. They also partner with the Software & Controls Development to build in the required device-level standardization required to capture into a standard module. The long-term vision is that this role develops the foundational blocks to enable seamlessly modular designs up-front with known costs, controls, and integration minimizing ambiguity for vendors and firm engineering.

4. Team Scope and Deliverables

Software & Controls Development Scope & Deliverables: The primary scope of the team is to focus on first-mile conveyance and sorter applications and excludes any equipment owned by FTR today. The solutions should be designed to be compatible with retrofitting a legacy network and new retrofit equipment projects, but can be used to partner with WWDE or One-MHS for future greenfield applications also. The team will deliver packaged solutions that are well-documented with vendor/Amazon scope clearly defined and internal execution requirements and processes fully developed. The solutions will be vendor agnostic and building type agnostic and will enable software orchestration and business level decision making to interact with while keeping flexibility for RME troubleshooting.

Alpha/Beta deployments are in-scope, but once graduated to general availability will be passed off to Software & Controls execution within the firm engineering team or in collaboration with vendors.

Software & Controls Design Scope & Deliverables: The primary scope of the team is the same projects that retrofit design team will be working on or software/controls retrofit-only projects. The team will evaluate existing hardware/controls configurations, interlocks, safety zones, and infrastructure requirements and document the retrofit integration requirements to be part of the RFP. Where applicable, the team will incorporate internally developed software solutions to be included into the design. Flow control management documentation will be developed as part of the RFP, in partnership with Retrofit design, detailing decision tree diagrams that enable a vendor to accurately quote and program logic meeting Amazon's expectation and functional requirements. Conceptually feasible software functional specifications and description of operations will be developed internally as opposed to the vendor. The team will align on internal requirements with clear scope delineation for vendors and drive partner team OP planning for software solutions that need to be developed to support the retrofits including sortation, OEE, and SCADA. Device layouts, panel design, load calculations, and line drawings are considered out of scope and part of vendor deliverables to be reviewed during firm engineering. Simulation is considered out of scope, but could be reevaluated particularly as standard code is developed.

Retrofit Design Scope & Deliverables: The primary scope of the team is to develop conceptual packaged solutions based on the intake from the planning team to align towards a concept and generate an RFP. The team will maximize integration of internalized standard modules and application standards where possible. They will evaluate internal and external technology alternatives, generate cost estimates, and assess lead times from either partnering with vendors or through a central PLM and cost estimation tool. The team is responsible to consider and document space constraints, equipment specifications, rate models, throughput and storage estimates, labor estimates, and infrastructure requirements (e.g. air, power, data) to generate thorough RFPs, facilitate informed decisions, and calculate accurate ROIs. They drive alignment of concepts and stakeholder reviews through LTM3/7/8 documents and reviews. Though the team is a key stakeholder in generating an RFP, they do not put together the final RFP nor own the RFP process. Requirements are documented and considered during concept design to ensure work is appropriately captured Pre-RFP, but compliance within the design (egress, safety, structural, etc.) is considered Firm Engineering scope. Concept Engineering for POR projects first mile and middle mile projects are considered out of scope, but standard mechanisms and workflows will be collaborated across multiple orgs.

Open Scope to be Resolved:

Onboarding and training of vendors to do software design integration (OEE, SCADA, ANOW, MOC) – TVM?

Development and deployment of vendor-based equipment data ingestion (Alarms, MySPD) – Geist, Firm?

Sortation service deployments – Firm, Field? Could own Autodeploy SOP and simplify for field?

IPC Provisioning – Firm, RME, IT?

NPI (IBIS, ARSAW) – Concept until graduated to firm?

FAT Testing – Are we a stakeholder? Probably only for Alpha/Beta not GA products

5. Mechanisms

The team uses SmartSheets and White Papers as the core mechanisms to ensure project tracking, stakeholder approvals, and design quality.

SmartSheets are used to track project schedule, risks, and communications. Each designer updates their project information on a weekly basis and it is reviewed in a team meeting. A monthly flash is then sent sharing projects at risk, projects behind SLA's, and critical design project highlights. Escalations are highlighted during the L8 WBR.

Once a designer receives a project, they review the business justification white paper to ensure the data justifies payback and that the design issue is well defined. This information will then be incorporated into the concept development White Paper review, which is an LTM3. Prior to CAR approval, the designer will write an LTM8 to gain final design direction. This process can be further reviewed in Appendix D.

Lessons learned will be generated through org wide mechanisms such as participation in Tiger Teams and MHE lifecycle management. To reinforce strong engineering development and learnings, designers will have audit gates to have structured touchpoints with firm engineering and field execution teams.

Other mechanisms include:

1. RFP review calls
2. Proposal Review calls with vendor selection matrix
3. Firm Engineering CO approvals
4. Proposal Checklists
5. Hand Off Document Templates

New development activities are tracked through SIMs and internal standard sprint planning tasks. Technical alignment and reviews are completed through PRFAQs, BRDs, and Technical Design docs. Code development uses git-based CI/CD frameworks such as code.amazon and Copia. Module development, standards, equipment catalogs, and cross-functional standards will require a unified repository such as a PLM (Product Lifecycle Management) tool.

6. Measures of Success

Speed & Efficiency Metrics

1. **Concept to Order Entry Cycle Time** – Measures the true design cycle time and how well early concept efforts reduce downstream engineering and vendor effort.
2. **Concept to RFP Cycle Time** – Ensures concept deliverables enable an effective RFP within SLA.
3. **SLA Adherence %** – Tracks milestone adherence to pinpoint bottlenecks in the concept-to-RFP process.
4. **Proposal Iteration Count** [Future] – Measures the number of iterations before a concept is finalized to identify inefficiencies.

Cost & Value Metrics

5. **Cost Estimate vs. Actual Variance %** – Measures estimation accuracy against final project costs to improve forecasting.
6. **Projected vs. Actual Benefits (Space, Throughput, Cube, etc.)** – Validates whether expected improvements materialized post-implementation.
7. **Conceptual Cost Reduction %** [Future]– Tracks cost savings driven by concept engineering before final design.
8. **Value Engineering Metrics (Cost per Linear Foot, Cost per Cubic Foot, etc.)** [Future] – Benchmarks efficiency of design against cost.

Quality & Design Robustness

9. **Concept to Implementation Success %** – Measures how often initial concepts translate into final implemented designs without project cancellations
10. **Concept Miss Root-Caused Change Orders** – Measures how often a concept requires change orders in firm engineering or execution due to initial oversight.
11. **Design Reuse Rate** [Future] – Tracks how often past concept designs or standard modules are leveraged in new projects to improve efficiency
12. **Firm Engineering Rework %** [Future] – Identifies how much concept work must be revised post-firm engineering to detect misalignment.
13. **Module Quality Score** [Future] – Measures performance of modular solutions based on standard quality KPIs (HIE, Jam DPMO, OEE Tech, MTTR, etc.)

Innovation & Strategic Growth

13. **Innovation Index** [Future] – Captures the planned integration of new technologies, automation, or software solutions in concepts.
14. **Sustainability Index** [Future]– Measures the inclusion of energy-efficient, low-waste, or green solutions in concept designs.
15. **Proactive Design %** [Future]– Measures the share of conceptual projects/whitepapers that are generated internally proactively vs. reactively as requested by the business

7. Customers and Partners

Primary customers are the Operations business units including ACES, Programs, Supply Chain, and network Operations teams. The team partners with these groups closely to define business strategy, understand data justification, review concepts, and gain approval of design direction.

Secondary customers will include firm engineering, ops engineering, RME, and WHS. Firm engineering and ops engineering are downstream customers who will require a documented hand off and a deep level of understanding of the design. This is critical due to ensure project continuity. While Operations is the main customer approving the design, RME and WHS are critical in the approval process to ensure safe and reliable equipment. They will be consulted through the White Paper process.

Key partners are One-MHS, AFT, AIT, WWDE, and GPO. These are key partnerships where defining swim lanes and aligning on strategic priorities will be important to the continued growth of Amazon. One-MHS and AIT will be key partners in aligning towards an industrial automation strategy where AFT, UTR, Sort Tech will be important to gather requirements from. WWDE will be a key partner to collaborate standardized designs with and to continue to support one another in strategic designs. GPO and TVM are key partners throughout the RFP process and to ensure proper vendors are evaluated for projects.

8. Appendix

Appendix A: Concept Team SLA's in Business Days

Milestone	Accountable Group	T Shirt Size			
		XSmall and Small	Medium	Large	XLarge
Evaluation	RESS	10	15	20	20
RFC Writing	Concept	3	4	5	5
RFC to ROM	Vendor	11	14	17	17
ROM to RFP	Concept	15	20	20	20
RFP to Proposal	Vendor	12	16	24	25
Proposal to Acceptance	Concept/Vendor	20	20	20	20
Proposal to PO	RESS/GPO	60	60	60	60
PO to FA	Vendor/FIRE	100	150	170	185

Appendix B: Project T-Shirt Sizes

Size	CapEx

X Small	<$250K
Small	$250K-$750K
Medium	$750K-$1.5M
Large	$1.5M-$5M
X Large	>$5M

Appendix C: First Mile Concept Design Team 3-year Plan

CANES 3YP 2024 July.docx

Appendix D: White Paper Review Process

Solutions Eng Leadership Review Process.docx

Appendix E: Implementation Path Forward

The First Mile Concept Design Team has been established for 3 years and has adapted to consistent changes. Through the changes, the team has kept consistency with experienced L6 talent, project tracking, and RFP/Proposal Reviews. With the current broader organizational changes, the scope and mechanisms will not change significantly. However, with some turnover, there are new blue badge and contractors being brought to the team. An onboarding plan and mentorship will be established based on past plans used for the team. For the two open roles, it will be critical to hire experienced talent that have extensive MHE knowledge to support team ramp up. Along with personnel changes, the team will have a new manager and leader, which could affect morale and employee engagement. The manager will conduct hand off discussions with the previous manager to ensure continuity and set clear expectations of each employee in a team setting.

In addition to personnel risks, the team will need to ensure best practices are adopted amongst upstream and downstream teams. Previously, the team worked with RESS and FIRE teams closely to establish the RPDP (Retrofit Product Development Process). These teams have not changed significantly, but additional support will be added such as controls and software integration. Further, the larger org now has middle mile firm engineering. The team will not support concepts for middle mile firm engineering, but an opportunity now exists to standardize design processes. The teams will work closely to ensure mechanisms and KPI's are consistent and impactful.

Appendix F: 2025 Goals

Quality: X% YoY improvement in Integrated Building Launch Quality Score

Safety: X% YoY reduction in serious incident rate and lost time incident rate

People: 95%+ cross-training in building type process knowledge

Schedules: 95%+ on-time delivery of critical path milestone SLAs

Cost: 10% improvement in project vintage across GES supported businesses through value engineering efforts